

## 2011 NDeRC Status Report



**NANOweek introduction of Scanning Tunneling Microscopy at Swanson Primary Center: students taking data (left); building a model (center); deploying the model using STM instrumentation with ND grad students (right.)**

For the past 13 years the Notre Dame QuarkNet Center has incorporated K-12 teachers and students into the life and work of the research community in high energy particle physics. Six years ago, the Notre Dame extended Research Community ([NDeRC](#)) was designed to increase the surface area of this distinctively Notre Dame approach to integrating STEM community. Three colleges, six departments, 14 T&R faculty and 19 graduate students have participated in extending the invitation to some 325 teachers and more than 18,000 students in Michiana. Now entering its fifth and final year of NSF GK-12 funding, NDeRC is establishing a hot-pluggable infrastructure for community engagement across the STEM disciplines.

In the 2010 – 2011 academic year, NDeRC expanded its model of week-long STEM Institutes for teachers and additional week-long classroom activities (STEMweeks) for students into a number of areas beyond our particle physics roots. Designed by Notre Dame faculty, graduate students and K-12 master teachers, these activities—most with versions across the full K-12 spectrum—are focal points for K-12/University collaborations, each supported by its own blog and wiki.

| NDeRC K-12 Classroom Activities 2010 – 2011 |                               |            |            |
|---------------------------------------------|-------------------------------|------------|------------|
| Subdiscipline                               | Collaboration                 | # Teachers | # Students |
| developmental biology and genetics          | <a href="#">BioEYES</a>       | 62         | 4020       |
| nanoscience and technology                  | <a href="#">NANO</a>          | 14         | 1006       |
| environmental engineering                   | <a href="#">ENVIRO</a>        | 11         | 977        |
| molecular biology                           | <a href="#">GENO</a>          | 4          | 198        |
| astrophysics                                | <a href="#">ASTRO</a>         | 4          | 180        |
| particle physics                            | <a href="#">CMS Dataclass</a> | 2          | 130        |
| computational science                       | <a href="#">Alice/MATLAB</a>  | 2          | 50         |

**In the past year alone, NDeRC STEMweeks have been in 260 different classrooms with 100 Michiana teachers and 6600 students in 70 different schools.**

NDeRC is establishing a network of online connections using social media to strengthen the ties knit through our face-to-face interactions. At our 2011 Collaborating for Research and Education Forum, 100 Michiana K-12 teachers subscribed to our [MichianaSTEM Community blog](#) (now at 163 total subscriptions.) We have established a database of some 325 Michiana K-12 STEM teachers who have collaborated with NDeRC in one or more activities; our [Michianastem.org](#) web of more than 50 wikis has grown to a membership of more than 400 members.

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